



A virtual boundary superimposed on Google Earth map using geofencing (Source: www.aisc.aero)

ly passive. However, they do not work very well in noisy environment, have very short range (max. 300-500m), and are mostly used along with other detection techniques.

Optical/infrared sensors. These sensors are essentially daylight or infrared (IR) scanners, which detect objects based on their appearance and motion features across consecutive frames. IR scanners are useful in conditions of low visibility.

However, there are disadvantages, such as low spatial resolution, low image contrast due to diffused edges, presence of noise and pulse disturbances, high false-alarm rates, relatively poor performance in dark, fog, etc, and are not preferred as stand-alone systems. Visuals of the drones and their payload provide useful forensic evidence.

Thermal detectors. The motors, batteries, and other on-board equipment of drones radiate significant amount of heat that imparts thermal signatures, which can be recognised by thermal sensors. Thermal detection has advantages in terms of weather resilience, identification availability, and lower costs, but the range is limited.

Hybrid detectors. Using a single method for detection may cause blind spots, making successful neutralisation difficult. Hybrid detection systems

with sensor fusion technology and joint control system provide greater accuracy and installation flexibility.

Radar and optical or thermal cameras provide good balance as radars scan the target area, while vision systems control the external and internal parameters, enabling accurate investigation using

image zoom, tilt, and focus. While the RF scanners can gather information like type, control commands, etc, they are unable to provide the location, for which they are combined with acoustic or optical/thermal sensors.

Countermeasure equipment

Countermeasure equipment physically destroys the drones by neutralising them or overpowering their movement, or even taking control of them. They are classified as destructive and non-destructive. Non-destructive countermeasures prevent secondary

crises-landing or crashing.

Drone jamming. Drone jamming involves paralysing radio communication between the target drone and its controller to make it uncontrollable by using powerful interfering RF signals. The options include jamming the drone-controller link or jamming the GPS link so that it loses control of 'auto-home' facility.

The jamming systems can be directional or omnidirectional. In directional jamming, the power is confined spatially, decreasing the possibility of interference with the co-located RF devices/equipment. Stationary jammers are installed at a fixed location, whereas mobile jammers operate from portable devices that are handheld or vehicle-mounted.

RF jammers are of medium cost and provide non-destructive neutralisation, have short range, may affect and/or jam other radio communications, and can result in unpredictable drone behaviour, such as unintentionally sending the drone to its target. They can cause significant unintended impacts, such as interference with the TV broadcasts, telecommunications, or even the air-traffic control systems.

GPS spoofers. Controllers use GPS satellites to navigate the drones. Fake GPS signals are used to spoof the communication with the satellites, preventing the drone from moving as intended by the controller. The drone is 'spoofed' into thinking it's somewhere else and loses control of auto-home facility and can be hijacked and diverted to a desired zone. Drone hijacking is an ideal approach, facilitating follow-up investigation.

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Geofencing. Geofence is a dynamically generated virtual perimeter for a real-world geographic area, which could be in a radius around a



A 3D radar based drone detection system (Source: www.spotterrf.com)